

Alexander Sanchez Cyril

Software Engineer · Full-Stack · AI & Agentic Systems

alexander.s.cyril@gmail.com | +234 902 216 2214 | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience designing and delivering production-grade software across full-stack development, backend engineering, distributed systems, and AI-powered applications. Skilled in building scalable APIs, cloud-native services, event-driven architectures, and intelligent agent workflows using technologies such as LangGraph, LangChain, LlamaIndex, and Retrieval-Augmented Generation (RAG). Experienced with containerization, CI/CD automation, Infrastructure as Code (Terraform), and modern software engineering practices. Proven ability to take systems from architecture and implementation through deployment and production operations, with a strong focus on scalability, reliability, maintainability, and long-term system quality.

TECHNICAL SKILLS

Languages

Python · JavaScript · TypeScript · C# · Java · Go

Backend Frameworks & APIs

ASP.NET Core · Spring Boot · Gin · Django · FastAPI · Express · Fastify

Frontend

React.js · Next.js · Tailwind CSS · HTML5 · CSS3 · Bootstrap

AI & Agentic Systems

LangGraph · LangChain · LangSmith · LlamaIndex · RAG · MCP · Vector Databases · OpenAI API · Multi-Agent Systems · Prompt Engineering

Databases

PostgreSQL · MongoDB · MySQL · SQLite · Vector DBs · SQL Design & Optimization

Cloud & DevOps

Docker · Docker Compose · GitHub Actions · CI/CD · AWS · Vercel · Render · Neon · API Monitoring Systems

Infrastructure as Code

Terraform · OpenTofu · AWS CloudFormation · Pulumi

Backend & Distributed Systems Engineering

Git · System Design · Clean Architecture · Microservices · Event-Driven Systems · REST APIs · API Design · JWT Authentication · RabbitMQ · Apache Kafka · Redis

PROFESSIONAL EXPERIENCE

Self-Employed | April 2023 – Present

Built and deployed multiple full-stack, backend, AI, and infrastructure-oriented applications. Responsible for full lifecycle engineering including architecture, development, deployment, and maintenance.

EL-Exousia Weather Intel — Agentic Weather Intelligence Platform

Live AI Project

Stack: *FastAPI · React · LangGraph · LangChain · LangSmith · LlamaIndex · MCP*

- Production-ready Agentic RAG system with autonomous multi-tool LLM orchestration
- Implemented LangGraph-based agent pipeline with context memory via vector retrieval
- Integrated live weather API tool-calling within a real-time reasoning pipeline
- Designed multi-step reasoning workflows using LangGraph orchestration

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AxioQuan — E-Learning & Marketplace Platform

Live Project

Stack: *Java · C# · Fastify (TypeScript) · Next.js*

- Full-featured online learning platform with course creation, quiz builder, and paid content marketplace
- Implemented role-based access control supporting student, instructor, and admin roles
- Built real-time messaging system between students and instructors
- Integrated payment processing for paid course enrollment
- Built backend APIs and database architecture for scalable content delivery

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PingRobot — URL Monitoring & Wake-Up Service

Live Project

Stack: *Next.js (Full-Stack) · TypeScript*

- Built a production monitoring tool that keeps free-tier cloud services (Render, Heroku, Railway, Fly.io) awake by pinging on a schedule
- Implemented aggressive health-check with 60-second timeout
- Designed retry mechanism with automatic recovery attempts (up to 3 retries) and intelligent wake detection to distinguish cold-start vs. live responses
- Exposes a consumable JSON API endpoint secured with API keys testable via Postman or any HTTP client
- Includes JSON viewer UI and Google/GitHub OAuth authentication for user account management
- Developed secure API key authentication system for controlled access

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SchemaLence — PostgreSQL Schema Explorer & Database Analysis Tool

Live Project

Stack: *Python · PostgreSQL · Tkinter · TTKBootstrap · Psycopg2 · Pandas*

- Developed a desktop database exploration tool for PostgreSQL schema inspection, documentation, and data analysis
- Implemented comprehensive schema discovery including tables, columns, primary keys, foreign keys, constraints, indexes, triggers, and ENUM definitions
- Built interactive data browsing capabilities with sortable table views and support for viewing live database records
- Added CSV export, clipboard integration, and automated schema documentation generation for improved developer productivity
- Designed a modern desktop interface using Tkinter and TTKBootstrap for efficient database navigation and visualization

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Universal Time — World Clock & Timezone API Platform

Live Project

Stack: *TypeScript · Express.js · Next.js*

- Built REST API for global timezone and real-time clock synchronization
- Developed frontend visualization for analog and digital time displays
- Containerized backend using Docker for consistent deployments

- Integrated API with frontend for real-time data rendering
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LEADERSHIP & ACADEMIC PROJECT EXPERIENCE

- Led multiple software development teams across university projects, coordinating planning, task allocation, implementation, testing, and final project delivery.
- Facilitated collaboration among team members by organizing development activities, resolving technical challenges, and ensuring project milestones were completed on schedule.
- Guided project architecture discussions, technology selection, and implementation strategies while promoting effective communication, accountability, and teamwork.
- Managed project timelines, integration efforts, and technical decision-making to ensure successful delivery of team-based software engineering projects.
- Applied software engineering best practices including requirements analysis, system design, version control, testing, documentation, and project planning.
- Collaborated with diverse team members in agile-style environments, strengthening leadership, problem-solving, project coordination, and software engineering practices.
- Demonstrated ownership and technical leadership by supporting teammates and helping drive projects from requirements gathering through final presentation and evaluation.
- Consistently contributed to high-quality project outcomes while fostering a collaborative and results-oriented team environment.

EDUCATION

Bachelor of Science in Software Development
Brigham Young University – Idaho

Graduated 2026

Associate of Applied Science in Software Development
Brigham Young University – Idaho

Graduated 2026

CERTIFICATIONS

Software Development · Brigham Young University – Idaho	2026
Web Development · Brigham Young University – Idaho	2025
Web & Computer Programming · Brigham Young University – Idaho	2024